Semiconductor Physics – Computational Exercise

Carrier statistics and the pn junction

Learning goals. After this exercise you should be able to

- calculate the free carrier densities n and p of a bulk semiconductor from the density of states and the occupation probability,
- explain the connection between Fermi energy, free carrier densities and doping densities, and go from any of them to the others,
- judge when the Maxwell–Boltzmann approximation holds and where it fails,
- put everything together and simulate a pn diode in thermal equilibrium by solving the Poisson equation.

Material. The Python file `exercise_carrier_statistics_pn_junction.py` contains a class for silicon with all important material parameters and some given functions, followed by one code cell (`# %%`) per task. Read the given parts before you start:

`Si(N_D, N_A, T=300)`	silicon with donor density N_D and acceptor density N_A in m^{-3}; contains E_C, E_V, N_C, N_V, n_i, ε_r and the energy grid `mat.E`
`calcDensityOfStates(E, mat, band)`	density of states $Z(E)$ of the conduction band (`'CB'`) or the valence band (`'VB'`)
`calcProbabilityDensity_FD(E, mat, Ef)`	Fermi–Dirac occupation probability $f(E)$
`calcEf_Exact(mat)`, `calcEf_Approx(mat)`	Fermi energy from the doping (given in Task 2b)
`Poisson(material, L)`	Poisson solver for a device of length L (given in Task 4, with two gaps)

Model. Silicon at $T = 300\,\mathrm{K}$, band gap $E_g = 1.12\,\mathrm{eV}$. The conduction band edge is the energy zero, $E_C = 0$, $E_V = -E_g$. Density-of-states masses $m^*_{\mathrm{dos},n} = 1.08\,m_e$ (six valleys) and $m^*_{\mathrm{dos},p} = 0.55\,m_e$ (heavy and light holes), $\varepsilon_r = 11.7$. With these masses, $N_C = 2.83 \times 10^{19}\,\mathrm{cm}^{-3}$, $N_V = 1.02 \times 10^{19}\,\mathrm{cm}^{-3}$ and $n_i = \sqrt{N_C N_V}\, e^{-E_g/2kT} = 6.7 \times 10^9\,\mathrm{cm}^{-3}$. All dopants are ionised. *The code uses SI units*: energies in J, densities in m^{-3}, lengths in m. Convert for the plots: energies in eV, densities in cm^{-3} ($1\,\mathrm{m}^{-3} = 10^{-6}\,\mathrm{cm}^{-3}$).

Hand-in. Your Python file `exercise_carrier_statistics_pn_junction.py` with *all* `TODO`s filled in. Running it must produce all plots (axis labels with units, legends). Answer the questions in *italics* as comments in the `TODO answer` fields of the corresponding cell. Two or three sentences per question are enough.

Task 1 Carrier densities from the Fermi energy

The processes in a semiconductor, like current transport, depend strongly on how the carriers are distributed over the conduction band (CB) and the valence band (VB). In this task you calculate the volume density of electrons and holes in a piece of silicon for a given Fermi energy.

Carrier densities

The electron density in the CB and the hole density in the VB are

$$n = \int_{E_C}^{\infty} Z_C(E)\, f(E)\, \mathrm{d}E, \qquad p = \int_{-\infty}^{E_V} Z_V(E)\, \left[1 - f(E)\right] \mathrm{d}E, \qquad f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}.$$

For $E - E_F \gg kT$, the Fermi–Dirac distribution can be replaced by the Maxwell–Boltzmann distribution $f(E) \approx e^{-(E-E_F)/kT}$. This gives $n \approx N_C\, e^{-(E_C-E_F)/kT}$ and $p \approx N_V\, e^{-(E_F-E_V)/kT}$ (lecture notes, chapter 3).

(a) Densities for a given Fermi energy. First plot $Z(E)$ of both bands and $f(E)$ for a few Fermi energies to understand the given functions. Then complete `calcElectronDensity_FD` and `calcHoleDensity_FD`. Integrate numerically on the energy grid `mat.E`, e.g. with `scipy.integrate.trapezoid`. Test them with a Fermi energy close to the CB ($E_F = E_C - 0.02\,\mathrm{eV}$, you should get $n = 1.13 \times 10^{19}\,\mathrm{cm}^{-3}$) and close to the VB ($E_F = E_V + 0.02\,\mathrm{eV}$).

Numerical hint: do not compute the hole occupation as `1 - f`. Where f is very close to 1, the difference is rounded to exactly 0 and p drops to zero. Use $1 - f(E) = 1/(1 + e^{(E_F - E)/kT})$ instead.

(b) Densities over the Fermi energy. Calculate n and p for Fermi energies from $E_V - 0.3\,\mathrm{eV}$ to $E_C + 0.3\,\mathrm{eV}$ and plot them over E_F. Use a logarithmic y-axis and indicate the band gap. *Where do the curves of n and p cross? What is this point called?*

(c) Maxwell–Boltzmann. Complete `calcProbabilityDensity_MB` and use it to calculate n and p in the Maxwell–Boltzmann approximation. Add the results to the plot of (b). Find the Fermi energy (relative to E_C, in units of kT) from which the Maxwell–Boltzmann result is more than 10 % too large. *Analyse the differences: where and why does the approximation fail? Which electron density does your limit correspond to? Compare with the statement in the lecture notes that silicon is non-degenerate below about* $10^{18}\,\mathrm{cm}^{-3}$.

Task 2 Fermi energy from the doping

Now we go the other way round. We know the doping of a piece of silicon and want to know the free carrier densities and the corresponding Fermi energy.

Doped semiconductor in equilibrium

With all dopants ionised, the free carrier densities follow from charge neutrality, $p - n + N_D - N_A = 0$, and the mass action law, $n\,p = n_i^2$ (lecture notes, chapter 3, "Charge carrier concentration in the doped semiconductor"). From n (or p), the Fermi energy is found by solving $n(E_F) = n$ with the exact Fermi–Dirac integral of Task 1. In the non-degenerate case with $N_D \gg N_A$ (or $N_A \gg N_D$), it simplifies to

$$E_F \approx E_C - kT \ln \frac{N_C}{N_D - N_A} \qquad \text{or} \qquad E_F \approx E_V + kT \ln \frac{N_V}{N_A - N_D}.$$

(a) Free carriers from the doping. Complete `calcChargeCarrierDensitiesFromDoping` with the formulas from the lecture notes.

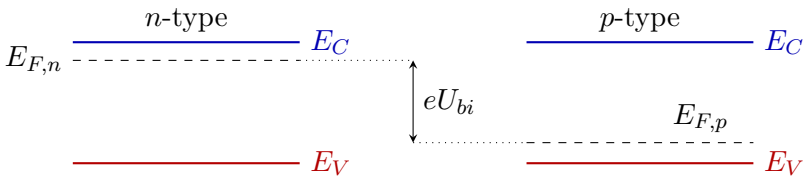
Numerical hint: for strong doping, n_i^2 is tiny compared with $(N_D - N_A)^2$. Calculate the majority carrier density first, and the minority carrier density from $n\,p = n_i^2$. A difference of two almost equal numbers would lose all significant digits.

(b) Fermi energy over doping. Two functions are given: `calcEf_Exact` solves $n(E_F) = n$ with the Fermi–Dirac integral (a root search with `scipy.optimize.bisect`), `calcEf_Approx` uses the formulas in the box. Plot E_F over the doping density with both functions, with a logarithmic doping axis. Make two plots. In the first one, keep the acceptors low ($N_A = 10^{12}\,\mathrm{m}^{-3}$) and vary the donors, e.g. $N_D = 10^{14} \ldots 10^{26}\,\mathrm{m}^{-3}$. In the second one, turn it around: $N_D = 10^{12}\,\mathrm{m}^{-3}$ and vary N_A. Mark E_C, E_V and the intrinsic level E_i. *What is the underlying approximation of `calcEf_Approx`? There is more than one. In which doping ranges does it fail, and what is the reason in each range?*

Now you have all functions to go from any starting point to any desired output. If you know the Fermi energy, you can calculate the carrier densities. If you know the doping, you can calculate the

free carriers and from them the Fermi energy. In the next steps, we use all of these connections to simulate a pn junction.

Task 3 Built-in voltage



Before contact, the n-type and the p-type piece each have their own Fermi energy. In contact and in thermal equilibrium, there is only one Fermi level in the whole device, so the bands of the two sides have to shift against each other by the difference of the Fermi energies. This difference is the built-in voltage, $e\,|U_{bi}| = E_{F,n} - E_{F,p}$.

Calculate the built-in voltage with `calcEf_Exact()`. First, check your result with the example of the lecture notes, $N_D = 10^{17}\,\mathrm{cm}^{-3}$ and $N_A = 5 \times 10^{16}\,\mathrm{cm}^{-3}$ (lecture notes: $U_{bi} \approx 0.836\,\mathrm{V}$). Then look up typical doping levels of a real pn diode and use those. That way, you can immediately cross-check the built-in voltage with real-world pn diodes. Remember that `Si(N_D, N_A)` takes the doping in m^{-3}. *Which doping levels did you use (and where did you find them)? Compare U_{bi} with the forward voltage of a real silicon diode. Can U_{bi} be larger than E_g/e? Try $N_D = N_A = 10^{20}\,\mathrm{cm}^{-3}$ and explain.*

Task 4 The pn junction in equilibrium: Poisson equation

Now we put everything together and simulate a pn junction in thermal equilibrium, where no current flows. This requires solving the Poisson equation. Its solution gives the potential across the whole device, including the transition region, without further approximations.

Discretised Poisson equation (lecture notes, chapter 3)

The electrostatic potential $V(x)$ shifts the band edges, $E_C(x) = E_C - eV(x)$, and satisfies

$$\varepsilon \frac{\mathrm{d}^2 V}{\mathrm{d}x^2} = -e\,(N_D - N_A - n + p).$$

On a grid x_i with spacing Δx, multiplied by $\Delta x^2/\varepsilon$, the residual at an inner grid point i is

$$R_i = V_{i-1} - 2V_i + V_{i+1} + \frac{e\,\Delta x^2}{\varepsilon}\,(N_D - N_A - n + p)_i, \qquad i = 1, \ldots, N-2.$$

`scipy.optimize.root` varies the inner potentials $V_1 \ldots V_{N-2}$ until all $R_i = 0$. The boundary values are fixed: $V(0) = 0$ and $V(L) = U_{bi}$, with $U_{bi} = (E_{F,\mathrm{right}} - E_{F,\mathrm{left}})/e$ from the neutral boundaries. The densities only depend on $E_F - E_C(x) = E_F + eV(x)$, so the code evaluates n and p at each point with the functions of Task 1 and the "shifted Fermi energy" $E_F + eV_i$.

(a) Complete the Poisson class. Make two modifications in the class `Poisson`:

1. In `__init__()`, calculate the built-in voltage and write it to the variable `V_built_in`. Mind the sign: the device has the n side on the left and the p side on the right.

2. In `poissonEquation()`, calculate the residuals R_i. The function must return the residuals at every grid point except the two boundary points, i.e. a vector of length $N-2$. Note that `n` and `p` are already given only at the inner points, while `N_D`, `N_A` and `self.poissonConst` are given at all N points.

(b) Simulate the diode. Run the example: $L = 3\,\mu\text{m}$, 200 grid points, $N_D = 10^{17}\,\text{cm}^{-3}$ on the left, $N_A = 5 \times 10^{16}\,\text{cm}^{-3}$ on the right. This takes about 20 s. Your results should look like the *pn* junction in the lecture notes (chapter 3, "Numerical solution of the Poisson equation"). *Compare the total potential drop with your U_{bi} from Task 3. How wide is the space-charge zone on the n side and on the p side, and why are they different? What is the total charge of the space-charge zone? The code takes E_F from the left boundary: what would change if you took it from the right boundary instead?*